

Enhancing 3D Cerebrovascular Segmentation: Integrating Dynamic Snake Convolution with Symmetric Equilibrium GANs for Thin Vessel Recovery

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Project Description

Background

Accurate 3D brain vessel segmentation is a critical prerequisite for advanced downstream medical modelling and quantification tasks. The precision of these segmentation mappings directly impacts the reliability of subsequent clinical analyses and diagnostic workflows. For example, high-quality segmentations can assist radiologists in locating and diagnosing lesions (Qi et al., 2023).

Problem Statement

Despite advancements in medical image processing, existing vessel segmentation pipelines consistently struggle to accurately identify thin, low-contrast tubular structures. While architectures like Dynamic Snake Convolution (DSConv) offer improvements over standard U-Net implementations by learning adaptive, curve-aware kernels, empirical observations indicate that DSConv still produces false negatives when processing highly elusive, thin, and low-contrast cerebrovascular vessels (Qi et al., 2023).

Proposed Solution

To address the persisting false negatives, this project proposes the integration of DSConv within a Symmetric Equilibrium Generative Adversarial Network (SEGAN) framework. SEGAN enhances the ability of conventional U-Net architectures to segment elusive thin vessels. It operates by pairing a U-Net generator with a U-Net discriminator, which generates symmetric adversarial pressure, effectively teaching the generator to reconstruct highly realistic local details that can recover the thin, low-contrast vessels typically missed by standalone U-Net models (Zhou et al., 2021).

Research Questions and Hypothesis

This research is guided by the following core questions:

- **RQ1:** Can the adversarial loss introduced by a SEGAN architecture reduce the false negative rate of DSConv on thin, low-contrast vessels?
 - **RQ1a:** Can this reduction in false negatives be achieved without simultaneously introducing hallucinated false positives?
 - **RQ1b:** Can the integration preserve the adaptive, curve-aware representations that the baseline DSConv inherently captures?

Hypothesis: While DSConv excels at learning adaptive, curve-aware kernels, it lacks the sensitivity required for thin, low-contrast vessels. Integrating DSConv into a SEGAN framework will force the model to prioritise these elusive structures through symmetric adversarial pressure, thereby solving the issue of false negatives without degrading existing geometric accuracy.

Novelty and Contributions

This project introduces two primary novel contributions to the field of medical image segmentation.

First, it adapts SEGAN—a framework traditionally restricted to 2D retinal image segmentation—for the complex, high-dimensional domain of 3D brain cerebrovascular volumes.

Second, it pioneers the integration of DSConv within a Generative Adversarial Network (GAN), leveraging adversarial pressure specifically as a mechanism to recover thin, low-contrast vascular structures.

Literature Review

Conventional 3D Vessel Segmentation Methods

Background and Problem Statement

Unlike bulk organ segmentation, isolating continuous networks like blood vessels and roads poses unique morphological and topological hurdles. Vessels and roads are characterised by their detailed structures, thin peripheral branches, and the presence of extremely small image components that are heavily influenced by surrounding noise and subtle variations in image intensity (Goni et al., 2022). In medical imaging, domain scientists frequently employ Maximum Intensity Projection (MIP) to suppress noise and observe vascular structures; however, automated segmentation still struggles.

A major spatial challenge arises from dimensionality. As Goni et al. (2022) point out, a vessel's direction extends across the X, Y, and Z axes. Applying 2D techniques to 3D volumetric images results in the loss of vital spatial context, while 3D convolutional neural networks (CNNs) require an enormous amount of computational time and parameters. Furthermore, complex vessel geometry, topological changes, and the sheer sparsity of vessel data in a large 3D volume create considerable obstacles (Goni et al., 2022). A persistent issue across domains is the failure to detect fine structures, such as missing thin terminal bronchioles or capillaries in final predictions, indicating that models still struggle with multiscale representation (Li et al., 2022).

Architecture

Historically, segmentation relied on multi-stage processes integrating anatomical knowledge, such as utilising skeletonisation-based algorithms to create atlases that guide mathematical morphology (Li et al., 2022). Today, Deep Learning (DL) has largely superseded these conventional methods due to its superior generalisation, recognition capabilities, and ability to automatically extract multi-level features without being constrained by handmade features (Li et al., 2022).

Recent reviews categorise these DL tasks into fully supervised, semi-supervised, and unsupervised learning (Liu et al., 2024). While fully supervised methods currently yield the highest precision, they demand massive amounts of annotated data. To address the computational bulk of advanced 3D models and Graph/Transformer-based approaches, researchers are exploring lightweight networks. Through techniques like knowledge distillation, crucial feature representations can be extracted from large models and transferred to smaller ones, significantly reducing computational complexity while maintaining accuracy (Liu et al., 2024; Goni et al., 2022). Additionally, attention mechanisms and Generative Adversarial Networks (GANs) have introduced new opportunities. GANs, in particular, are being leveraged to automatically generate data annotations and bridge the gap between synthetic and real imagery (Goni et al., 2022; Liu et al., 2024).

Limitations

Despite architectural advances, the primary bottlenecks in modern network segmentation revolve around data limitations and the metrics used to evaluate them.

Data Scarcity and Privacy

Deep learning models require vast datasets to learn effectively. In medical imaging, privacy regulations severely restrict data sharing. Goni et al. (2022) note that over 80% of reviewed BVS studies utilised private datasets, creating a "deadlock" for researchers and preventing the development of models that can generalise across different data distributions. Both Goni et al. and Li et al. agree that substantial, publicly available databases are necessary for the field to flourish.

Class Imbalance

Blood vessels typically constitute less than 3% of the voxels in a patient's 3D image volume (Goni et al., 2022). This extreme bias toward the background class causes existing class-balancing loss functions to become numerically unstable. It also exacerbates issues like gradient vanishing during backpropagation in deep networks.

Metric Shortcomings

Relying solely on standard metrics like Accuracy, Intersection over Union (IoU), and Dice coefficients is inadequate for spatial and topological validation. These metrics do not capture discontinuity in peripheral networks. Consequently, there is a push to incorporate connectivity-emphasising loss functions, such as cIDice, to ensure the topological integrity of the segmented networks (Li et al., 2022).

Future Directions

Looking forward, several key strategies have been identified to overcome current limitations:

Unsupervised and Semi-Supervised Learning

Transitioning away from fully supervised models reduces the reliance on large-scale annotated datasets. These methods lower data preparation costs and actively mine implicit structures within images, presenting a promising solution to the medical data scarcity dilemma (Liu et al., 2024; Li et al., 2022).

Multi-Modal Data Fusion

Integrating data from diverse sensors (e.g., combining different MRI sequences, or merging LiDAR with remote sensing imagery) addresses the limitations of a single data source and improves feature capture across varying environments (Liu et al., 2024).

Explainability

As deep neural networks act as "black boxes," their lack of interpretability remains a concern. Future research must prioritise explainable AI solutions through sensitivity analysis, gradient-based methods, and rigorous multi-centre data harmonisation (Li et al., 2022; Liu et al., 2024).

DSConv (Qi et al., 2023)

Background and Problem Statement

Conventional Convolutional Neural Networks (CNNs) have been widely applied to medical image segmentation, but they face a severe morphological bottleneck when extracting tubular structures like blood vessels. These structures are typically thin, tortuous, and account for a minute proportion of the overall image volume, making them highly susceptible to complex background interference. Historically, standard convolutional kernels are adept at extracting local features but fail to capture complex, curving global morphologies.

To adapt to geometric variations, deformable convolutions are frequently employed. However, in the context of fragile tubular structures, allowing the network completely free rein to learn geometric deformation creates a structural vulnerability: the receptive field frequently strays outside the target region and into the background. Consequently, the model struggles to differentiate the thin structure from the noise, resulting in fractured and discontinuous segmentations.

Architectural Innovations of DSCNet To resolve this morphological bottleneck, Qi et al. (2023) introduced the Dynamic Snake Convolution Network (DSCNet). DSCNet's primary innovation is the integration of domain knowledge regarding tubular morphology directly into the feature extraction and fusion pipelines. Instead of unconstrained learning, it supplements the free learning process with topological constraints. To support this architecture, DSCNet introduces three specialised components:

- **Dynamic Snake Convolution (DSConv):** Instead of allowing a free-roaming receptive field, DSConv straightens the convolution kernel and iteratively selects observation positions based on a cumulative offset. By forcing the kernel to conform to a linear, snake-like morphological structure, it adaptively focuses on slender and tortuous local features. This ensures the network's attention remains tightly bound to the vessel without wandering into surrounding tissue.
- **Multi-View Feature Fusion Strategy:** To capture complex and variable global morphologies, this module generates multiple morphological kernel templates based on the DSConv. By extracting features from both the x-axis and y-axis across varying

offsets, the network observes the structural characteristics of the target from multiple perspectives. It then summarises these vital standard features to ensure important information from diverse global orientations is retained.

- **Topological Continuity Constrained Loss (TCLoss):** To physically enforce connectivity, DSCNet utilises Persistent Homology (PH) to map the emergence and disappearance of topological features. By calculating the bidirectional Hausdorff distance between the predicted topology and the ground truth, TCLoss rigorously penalises anomalous outlier points. This exerts intense pressure on the network to focus on fracture regions with abnormally low voxel distributions.

Performance and Applicability

DSCNet is natively designed to process both 2D and 3D feature maps, making it highly versatile for volumetric cerebrovascular segmentation. Empirically, this constrained framework achieves state-of-the-art results on both 2D retinal datasets (DRIVE) and complex 3D volumes (Cardiac CCTA). On 3D networks, DSCNet paired with TCLoss demonstrates significant improvements in centerline-Dice (clDice) and Overlap until First error (OF) metrics. This demonstrates a distinct ability to preserve the continuity of minority-class pixels (thin vessels) in high-dimensional spaces without sacrificing the overall volumetric Accuracy or Specificity.

Limitations

While highly effective for preserving the topological integrity of thin vessels, scaling the DSCNet framework requires careful architectural and computational management. The Multi-View Feature Fusion strategy, by generating numerous morphological kernel templates, inevitably introduces redundant noise and expands the memory footprint. Consequently, it relies heavily on a random dropping strategy during training to mitigate overfitting. Furthermore, computing Persistent Homology and Hausdorff distances for the TCLoss is mathematically intensive and inherently sensitive to outliers. If the data contains high levels of noise, the loss landscape can become volatile. Balancing the cross-entropy loss with this topological constraint requires meticulous hyperparameter tuning to ensure the network heals fractured vessels without hallucinating false positive connections.

SEGAN (Zhou et al., 2021)

Background and Problem Statement

Generative Adversarial Networks (GANs) have been increasingly applied to medical image segmentation to improve the realistic reconstruction of anatomical structures. However, traditional adversarial training in this domain suffers from an asymmetric structural imbalance between the Generator (G) and the Discriminator (D). Historically, advanced architectures like U-Net are employed as the generator to map complex features, while the discriminator relies on a simpler, shallower classification network.

In the context of segmenting elusive, thin vessels (which are often only 2-3 pixels wide and obscured by low-contrast backgrounds or lesions), this asymmetry creates a performance bottleneck. If the discriminator is structurally weak, the adversarial evaluation is approximately 1 (high) even when the generator's output lacks fine-grained details. Consequently, the generator is not "pressured" to improve its segmentation of micro-vessels, as the discriminator cannot reliably differentiate between a perfect segmentation and an imperfect one.

Architectural Innovations of SEGAN To resolve this bottleneck, Zhou et al. (2021) introduced the Symmetric Equilibrium Generative Adversarial Network (SEGAN). SEGAN's primary innovation is the deployment of a full U-Net architecture for both the generator and the discriminator.

This symmetric configuration creates a "well-matched game" that empowers the discriminator to recognise pixel-level detail differences, thereby exerting maximum adversarial pressure on the generator to map highly realistic, high-resolution local details. To support this architecture, SEGAN integrates two specialised modules:

- **Multi-Scale Feature Refine Blocks (MSFRB):** Traditional U-Net up-sampling merges high-resolution features with high-semantic features using convolution, which often blurs and confuses high-resolution details. MSFRB introduces a dedicated side-branch that skips this convolutional confusion, squeezing adjacent channels to independently preserve high-resolution, low-semantic features.

- **Attention Mechanism (AM):** Embedded within the MSFRB, a channel-wise attention mechanism is utilised to adaptively allocate weights to discriminative feature channels, ensuring the network focuses heavily on informative vessel structures rather than background noise.

Performance and Applicability

SEGAN is trained using an optimised, composite objective function that balances standard adversarial loss with pixel-level Binary Cross-Entropy (BCE) and Mean Absolute Error (MAE) losses to smooth vessel edges and reduce false positives. Empirically, this symmetric framework achieves state-of-the-art results on 2D retinal datasets, notably yielding massive gains in Sensitivity (0.8294 on DRIVE, 0.8812 on STARE). This demonstrates a distinct ability to identify minority-class pixels (thin vessels) without sacrificing global specificity, alongside a highly efficient inference time of 0.16 seconds per image.

Limitations

While highly effective for 2D elusive vessel recovery, scaling SEGAN for 3D cerebrovascular segmentation presents challenges. Utilising a complex U-Net structure for both the generator and discriminator effectively doubles the parameter count and memory footprint compared to standard GAN configurations. Furthermore, SEGAN relies heavily on a composite loss function with delicate hyperparameters. Forcing a model to become hyper-sensitive to elusive structures risks generating hallucinated false positives if the loss landscape is not properly balanced. Translating this symmetric adversarial pressure to 3D volumes—especially alongside topology-aware modules like DSConv—will require rigorous memory management and loss function ablation to maintain geometric accuracy.

Progress Made so Far

Research Gap Identification and Refinement

A substantial portion of the initial project timeline was deliberately invested in independent direction scoping and gap analysis. While initially working with DSConv, an extensive review of existing image segmentation architectures for tubular structures was required to understand the current landscape.

This phase involved rigorously evaluating what had already been achieved in the field, analysing the strengths of current approaches, and critically identifying their remaining weaknesses. This extensive scoping process was necessary to ensure that the project would make a genuinely novel contribution.

Through this iterative refinement, the specific failure modes of DSConv were analysed, and the potential of adversarial networks was identified as a viable countermeasure. This scoping effort directly culminated in the formulation of the current problem statement, research questions, and proposed solution.

Ultimately, this phase successfully verified that the integration of DSConv within a GAN framework—specifically for 3D brain vessel segmentation—represents a highly valuable, unexplored intersection in current research, ensuring the project direction is both novel and rigorously validated.

DSConv Setup and Experimentation

Initial technical work focused on engineering the DSConv 3D volumetric data pipeline, encompassing data loading, preprocessing, and the training environment. To establish a pre-intervention performance ceiling, baseline training runs were executed using standard 3-layer, 4-layer, and 5-layer U-Net architectures using the MiniVess dataset (Poon et al., 2023). These initial runs served two direct purposes: they successfully validated the integrity of the computational environment and data scripts, and they generated the baseline

performance metrics required to accurately quantify the subsequent gains of the DSConv and SEGAN integrations.

Empirical analysis of the training logs revealed that all three architectural depths yielded remarkably similar terminal performance metrics. Specifically, the test sets plateaued at comparable minimum loss values, yielding average Dice Similarity Coefficients (DSC) of ~ 0.84 and centerline-Dice (clDice) scores of ~ 0.63 across the variants (Fig. 1).

Interestingly, observational data indicated that while the 4-layer configuration exhibited a marginally slower initial convergence rate, all variants ultimately required similar temporal scales to reach minimum loss (Fig. 2). Crucially, the tightly clustered, plateaued metrics across the 3, 4, and 5-layer configurations empirically validate that simply increasing the depth of standard convolutional architectures is insufficient for resolving the remaining structural drop-offs. This directly justifies the necessity of introducing topology-aware kernels (DSConv) and adversarial regularisation (SEGAN) to break past this established performance ceiling.

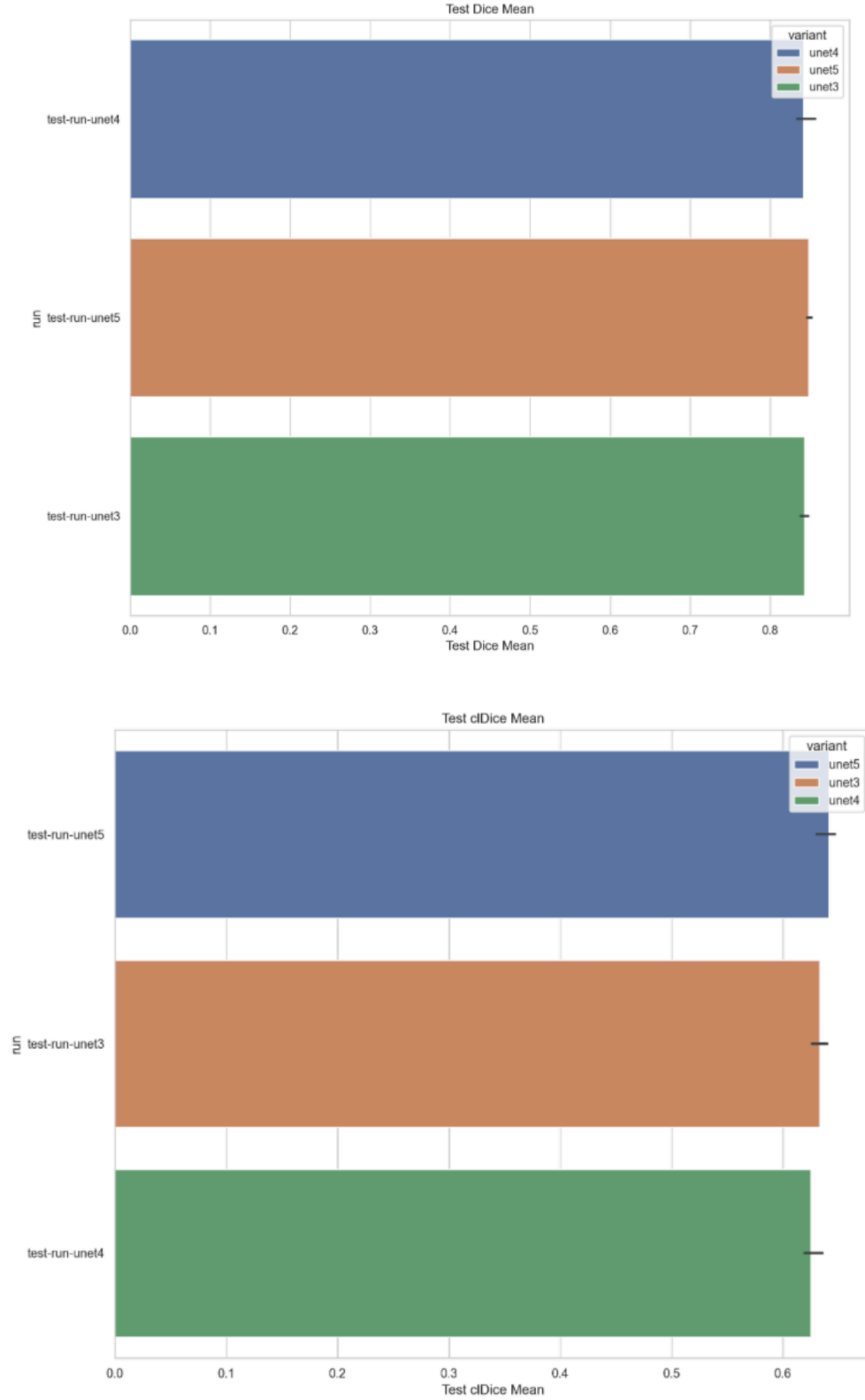


Figure 1: Empirical baseline test metrics (DSC and cDice) evaluated across 3-layer, 4-layer, and 5-layer U-Net configurations. The negligible variance in topological connectivity across network depths empirically demonstrates that simply deepening standard convolutional layers fails to resolve the structural discontinuity of thin vessels.

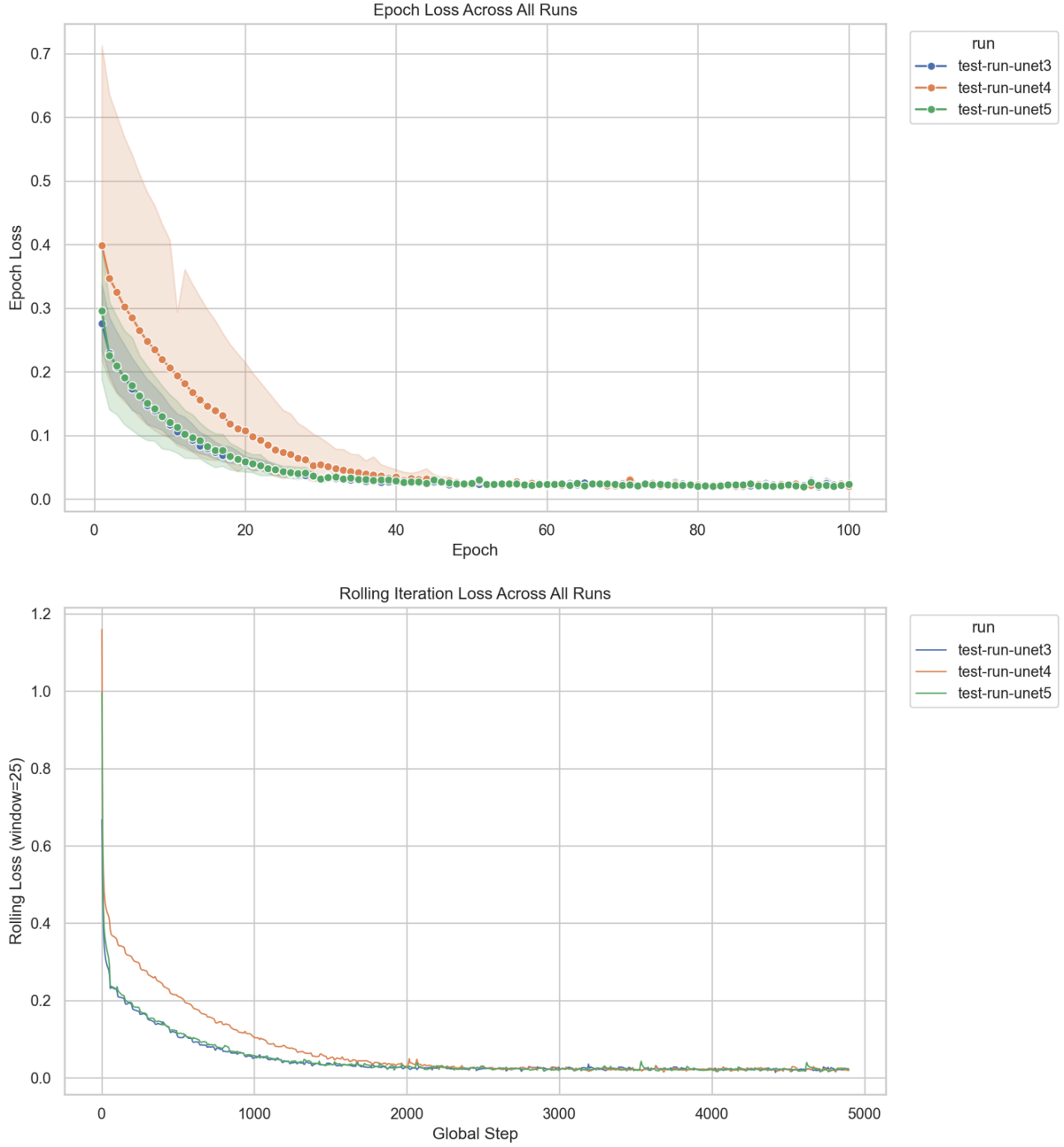


Figure 2: Training loss convergence plotted over epochs and rolling iterations for the baseline standard U-Net configurations. While the 4-layer variant demonstrated a marginally delayed initial convergence trajectory, all architectural depths successfully plateaued at equivalent terminal loss values within the same temporal training scale.

Research Plan for the Next Semester

Phased Implementation Strategy

The primary focus of the upcoming semester will be the technical integration of DSConv modules into the SEGAN framework. Given the known risks of exponential parameter expansion and convergence difficulties when extending architectures from 2D to 3D, the implementation will be structured into a rigorous, two-phase approach:

Phase 1: 2D Proof of Concept

Initial integration will utilise the 2D formulation of DSConv within the native 2D SEGAN architecture. This will allow us to experiment without the massive computational overhead of 3D volumes. Specifically, this phase targets the Generator (G). We will replace the standard U-Net Generator with a DSConv-equipped U-Net, as G is responsible for mapping the curve-aware tubular structures. The Discriminator (D) will initially remain a standard U-Net to establish a stable adversarial baseline and test loss functions. Using a 2D approach will allow us to test

Phase 2: 3D Extension

Following successful validation, the framework will be scaled to process 3D cerebrovascular datasets. To manage data scarcity and ensure robust topological validation, this scaling will utilize a multi-tiered dataset approach. Initial 3D integration will be trained on the DeepVesselNet dataset—a synthetic 3D vessel dataset providing perfect ground truth for thin structures (Tetteh et al., 2020). Following synthetic validation, the architecture will be tested against complex in vivo morphological structures using the MiniVess (Rodent) developed by Poon et al. (2023) and SYNAPSE (Marmoset) datasets. The 3D DSConv modules will be integrated into the 3D Generator (G). Because 3D convolutions inherently cause parameter ballooning, we will carefully monitor memory constraints.

Further Experimentation

If 3D segmentation is computationally viable, subsequent iterations will test other aspects as detailed in subsections below.

Further Experimentation

To ensure methodological rigour and accurately attribute performance gains to specific architectural decisions, the research will incorporate structured ablation studies and comprehensive loss function analyses:

Architectural Ablation Studies

- *Attention Mechanism Ablation:* We will train the network with and without the native SEGAN attention module to determine if its inclusion justifies the added parameter count, particularly when transitioning to 3D.
- *Generator vs. Discriminator Placement:* We will conduct ablation tests to compare a standard U-Net baseline against permutations where the DSConv modules are utilised exclusively in G, exclusively in D, and within both G and D simultaneously.

Loss Function Experimentation

Because GAN optimisation is notoriously volatile, particularly for thin structures, the loss landscape will be rigorously tested. We will expand beyond standard Binary Cross-Entropy (BCE) to experiment with topology-aware losses. Specifically, we will evaluate the **Topological Constraint Loss (TCLoss)** proposed in the original DSConv literature, as well as **Dice Loss** and **centerline-Dice (clDice) Loss**, to determine which mathematical constraint best enforces connectivity and continuity in thin, elusive vessels (Li et al., 2022).

Continuous Evaluation and Benchmarking

Rather than treating evaluation as a final step, performance benchmarking will be conducted continuously throughout the implementation and ablation phases. The evaluation of the proposed DSConv+SEGAN permutations against the baseline DSConv architecture will utilise a multi-faceted approach:

Quantitative Segmentation Metrics

To objectively measure the reduction in false negatives on thin vessels and overall structural continuity, we will compute standard **Dice Similarity Coefficient (DSC)**, Sensitivity, and Specificity, alongside topology-centric metrics like **clDice (centerline-Dice)**, which is specifically designed to evaluate the connectivity of tubular networks.

Computational Efficiency Profiling

Establishing a robust benchmark for architectural speed is imperative. We will assess total parameter count, training convergence speed, and inference time to quantify the exact trade-offs between segmentation accuracy and computational overhead as the model scales.

Qualitative Visualisation

Error maps will be generated to visually highlight missed vessels in the baseline model versus the recovered vessels in the integrated model, directly addressing the core research questions.

Proposed Gantt Chart

Week 1 is not necessarily Week 1 of next semester. We will start work in May.

Week	Phase 1: 2D Proof of Concept	Phase 2: 3D Volumetric Scaling	Architectural Ablation Studies	Loss Function Analysis
W1	Set up 2D DSConv in Generator; establish baseline.			
W2	Stabilise G/D adversarial equilibrium.			
W3	Finalise 2D validation and lock baseline.			
W4		Transition to 3D datasets; establish data pipeline.		
W5		Optimise 3D memory constraints and batch sizing.		
W6		Finalise 3D Generator scaling.	Begin Attention Mechanism ablation tests.	
W7		Test 3D expansion into Discriminator (if viable).	Compare G-only vs. D-only module placement.	

W8			Conclude G+D module ablation testing.	Initiate standard BCE vs. TCLoss experiments.
W9				Evaluate Dice and cLDice topological losses.
W10				Tune Discriminator adversarial pressure weights.
W11				Finalise optimal loss function weighting.
W12				
W13				

References

- Chen, H., Xiao, H., Liu, T., & Peng, H. (2015). SmartTracing: Self-learning-based Neuron reconstruction. *Brain Informatics*, 2(3), 135–144.
<https://doi.org/10.1007/s40708-015-0018-y>
- Goni, M. R., Ruhaiyem, N. I. R., Mustapha, M., Achuthan, A., & Che Mohd Nassir, C. M. N. (2022). Brain Vessel Segmentation Using Deep Learning—A Review. *IEEE Access*, 10, 111322–111336. <https://doi.org/10.1109/ACCESS.2022.3214987>
- Huang, Q., Hu, S., Liu, S., Ren, L., & Quan, T. (2026). Ultrafast 3D segmentation of brain-wide optical neuronal volume. *Biomedical Signal Processing and Control*, 111, 108324. <https://doi.org/10.1016/j.bspc.2025.108324>
- Hwu, Y., Margaritondo, G., & Chiang, A.-S. (2017). Q&A: Why use synchrotron x-ray tomography for multi-scale connectome mapping? *BMC Biology*, 15(1), 122.
<https://doi.org/10.1186/s12915-017-0461-8>
- Jiang, Z., Zhang, H., Wang, Y., & Ko, S.-B. (2018). Retinal blood vessel segmentation using fully convolutional network with transfer learning. *Computerized Medical Imaging and Graphics*, 68, 1–15.
<https://doi.org/10.1016/j.compmedimag.2018.04.005>
- Li, H., Tang, Z., Nan, Y., & Yang, G. (2022). Human treelike tubular structure segmentation: A comprehensive review and future perspectives. *Computers in Biology and Medicine*, 151, 106241.
<https://doi.org/10.1016/j.compbiomed.2022.106241>

- Liu, R., Wu, J., Lu, W., Miao, Q., Zhang, H., Liu, X., Lu, Z., & Li, L. (2024). A Review of Deep Learning-Based Methods for Road Extraction from High-Resolution Remote Sensing Images. *Remote Sensing*, 16(12), 2056.
<https://doi.org/10.3390/rs16122056>
- Poon, C., Teikari, P., Rachmadi, M. F., Skibbe, H., & Hynynen, K. (2023). A dataset of rodent cerebrovasculature from in vivo multiphoton fluorescence microscopy imaging. *Scientific Data*, 10(1), 141. <https://doi.org/10.1038/s41597-023-02048-8>
- Qi, Y., He, Y., Qi, X., Zhang, Y., & Yang, G. (2023). Dynamic Snake Convolution based on Topological Geometric Constraints for Tubular Structure Segmentation. 2023 *IEEE/CVF International Conference on Computer Vision (ICCV)*, 6047–6056.
<https://doi.org/10.1109/ICCV51070.2023.00558>
- Tetteh, G., Efremov, V., Forkert, N. D., Schneider, M., Kirschke, J., Weber, B., Zimmer, C., Piraud, M., & Menze, B. H. (2020). DeepVesselNet: Vessel Segmentation, Centerline Prediction, and Bifurcation Detection in 3-D Angiographic Volumes. *Frontiers in Neuroscience*, 14, 592352. <https://doi.org/10.3389/fnins.2020.592352>
- Xu, F., Shen, Y., Ding, L., Yang, C.-Y., Tan, H., Wang, H., Zhu, Q., Xu, R., Wu, F., Xiao, Y., Xu, C., Li, Q., Su, P., Zhang, L. I., Dong, H.-W., Desimone, R., Xu, F., Hu, X., Lau, P.-M., & Bi, G.-Q. (2021). High-throughput mapping of a whole rhesus monkey brain at micrometer resolution. *Nature Biotechnology*, 39(12), 1521–1528.
<https://doi.org/10.1038/s41587-021-00986-5>
- Xu, J., Chen, X., & Zhang, L. (n.d.). *Learning Dynamic Collaborative Network for Semi-supervised 3D Vessel Segmentation*.

Zhou, Y., Chen, Z., Shen, H., Zheng, X., Zhao, R., & Duan, X. (2021). A refined equilibrium generative adversarial network for retinal vessel segmentation. *Neurocomputing*, 437, 118–130. <https://doi.org/10.1016/j.neucom.2020.06.143>